CHAPTER 5 Thermodynamics

Thermodynamic Terms

1. Match List I with List II. *(2024)*

List I (Process)		List II (Conditions)	
A.	Isothermal process	I.	No heat exchange
B.	Isochoric process	II.	Carried out at constant temperature
C.	Isobaric process	III.	Carried out at constant volume
D.	Adiabatic process	IV.	Carried out at constant pressure

Choose the correct answer from the options given below:

a. A-I, B-II, C-III, D-IV b. A-II, B-III, C-IV, D-I
c. A-IV, B-III, C-II, D-I d. A-IV, B-II, C-III, D-I

Measurement of ΔU & ΔH, Work Done and Heat Capacity

2. Choose the correct statement for the work done in the expansion and heat absorbed or released when 5 litres of an ideal gas at 10 atmospheric pressure isothermally expands into vacuum until volume is 15 litres: (2024 Re)

a. Both the heat and work done will be greater than zero
b. Heat absorbed will be less than zero and work done will be positive
c. Work done will be zero and heat will also be zero
d. Work done will be greater than zero and heat will remain zero

3. For an endothermic reaction: *(2024 Re)*

A. q_p is negative. B. $\Delta_r H$ is positive.
C $\Delta_r H$ is negative. D. q_p is positive.

Choose the correct answer from the options given below:

a. B and D b. C and D c. A and B d. A and C

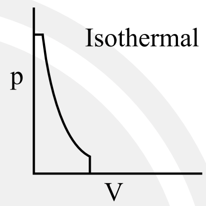
4. The work done during reversible isothermal expansion of one mole of hydrogen gas at 25 °C from pressure of 20 atmosphere to 10 atmosphere is: *(2024)*
(Given R = 2.0 cal K^{-1} mol^{-1})

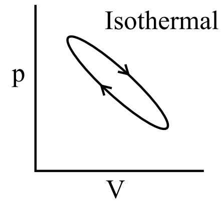
a. 413.14 calories b. 100 calories
c. 0 calorie d. – 413.14 calories

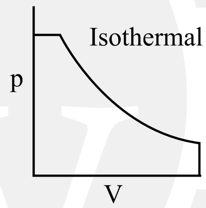
5. One mole of an ideal gas at 300 K is expanded isothermally from 1 L to 10 L volume. ΔU for this process is: *(2022 Re)*
(Use R = 8.314 J K^{-1} mol^{-1})

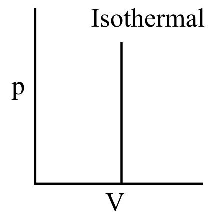
a. 0 J b. 1260 J c. 2520 J d. 5040 J

6. Which of the following p-V curve represents maximum work done? *(2022)*

a. 

b. 

c. 

d. 

7. Which one among the following is the correct option for right relationship between C_P and C_V for one mole of ideal gas? *(2021)*

a. $C_P - C_V = R$ b. $C_P = RC_V$
c. $C_V = RC_P$ d. $C_P + C_V = R$

8. The correct option for free expansion of an ideal gas under adiabatic condition is : *(2020)*

a. q = 0, ΔT < 0 and w > 0 b. q < 0, ΔT = 0 and w = 0
c. q > 0, ΔT > 0 and w > 0 d. q = 0, ΔT = 0 and w = 0

9. Under isothermal condition, a gas at 300 K expands from 0.1 L to 0.25 L against a constant external pressure of 2 bar. The work done by the gas is (Given that 1 L bar = 100 J) *(2019)*

a. –30 J b. 5 kJ c. 25 J d. 30 J

10. A gas is allowed to expand in a well insulated container against a constant external pressure of 2.5 atm from an initial volume of 2.50 L to a final volume of 4.50 L. The change in internal energy ΔU of the gas in joules will be: *(2017-Delhi)*

a. +505 J b. 1136.25 J c. –500 J d. –505 J

Enthalpy Change, ΔH of Reaction

11. For the following reaction at 300 K

$A_2(g) + 3B_2(g) \rightarrow 2AB_3(g)$

the enthalpy change is +15 kJ, then the internal energy change is : *(2024 Re)*

a. 19988.4 J b. 200 J
c. 1999 J d. 1.9988 kJ

12. Which amongst the following options is the correct relation between change in enthalpy and change in internal Energy? *(2023)*

a. $\Delta H + \Delta U = \Delta nR$
b. $\Delta H = \Delta U - \Delta n_g RT$
c. $\Delta H = \Delta U + \Delta n_g RT$
d. $\Delta H - \Delta U = -\Delta nRT$

Enthalpies For Different Types of Reactions

13. Consider the following reaction:- *(2023-Manipur)*

$2H_2(g) + O_2(g) \rightarrow 2H_2O(g) \Delta_r H^\circ = -483.64$ kJ.

What is the enthalpy change for decomposition of one mole of water? (Choose the **right** option).

a. 120.9 kJ
b. 241.82 kJ
c. 18 kJ
d. 100 kJ

14. At standard conditions, if the change in the enthalpy for the following reaction is –109 kJ mol^{-1}

$H_2(g) + Br_2(g) \rightarrow 2HBr(g)$

Given that bond energy of H_2 and Br_2 is 435 kJ mol^{-1} and 192 kJ mol^{-1}, respectively, what is the bond energy (in kJ mol^{-1}) of HBr? *(2020-Covid)*

a. 736
b. 518
c. 259
d. 368

15. The bond dissociation energies of X_2, Y_2 and XY are in the ratio of 1 : 0.5 : 1. ΔH for the formation of XY is –200 kJ mol^{-1} *(2018)*

The bond dissociation energy of X_2 will be

a. 200 kJ mol^{-1}
b. 100 kJ mol^{-1}
c. 400 kJ mol^{-1}
d. 800 kJ mol^{-1}

16. The heat of combustion of carbon to CO_2 is –393.5 kJ/mol. The heat released upon formation of 35.2 g of CO_2 from carbon and oxygen gas is: *(2015)*

a. + 315 kJ
b. – 630 kJ
c. – 3.15 kJ
d. – 315 kJ

Spontaneity, Entropy, Gibbs Energy Change and Equilibrium

17. In which of the following processes entropy increases?

A. A liquid evaporates to vapour *(2024)*

B. Temperature of a crystalline solid lowered from 130K to 0 K.

C. $2\ NaHCO_{3(s)} \rightarrow Na_2CO_{3(s)} + CO_{2(g)} + H_2O_{(g)}$

D. $Cl_{2(g)} \rightarrow 2\ Cl_{(g)}$

Choose the correct answer from the options given below:

a. A, C and D
b. C and D
c. A and C
d. A, B and D

18. For irreversible expansion of an ideal gas under isothermal condition, the correct option is: *(2021)*

a. $\Delta U \neq 0, \Delta S_{total} \neq 0$
b. $\Delta U = 0, \Delta S_{total} \neq 0$
c. $\Delta U \neq 0, \Delta S_{total} = 0$
d. $\Delta U = 0, \Delta S_{total} = 0$

19. For the reaction, $2Cl(g) \rightarrow Cl_2(g)$, the correct option is: *(2020)*

a. $\Delta_r H > 0$ and $\Delta_r S < 0$
b. $\Delta_r H < 0$ and $\Delta_r S < 0$
c. $\Delta_r H < 0$ and $\Delta_r S < 0$
d. $\Delta_r H > 0$ and $\Delta_r S > 0$

20. If for a certain reaction $\Delta_r H$ is 30 kJ mol^{-1} at 450 K, the value of $\Delta_r S$ (in JK^{-1} mol^{-1}) for which the same reaction will be spontaneous at the same temperature is *(2020-Covid)*

a. –33
b. 33
c. –70
d. 70

21. In which case change in entropy is negative? *(2019)*

a. Evaporation of water
b. Expansion of a gas at constant temperature
c. Sublimation of solid to gas
d. $2H(g) \rightarrow H_2(g)$

22. For a given reaction, ΔH = 35.5 KJ mol^{-1} and ΔS = 83.6 JK^{-1} mol^{-1}. The reaction is spontaneous at: (Assume that ΔH and ΔS do not vary with temperature) *(2017-Delhi)*

a. T > 298 K
b. T < 425 K
c. T > 425 K
d. All temperatures

23. For a sample of perfect gas when its pressure is changed isothermally from P_i to P_f, the entropy change is given by: *(2016 - II)*

a. $\Delta S = nRT \ln\left(\frac{P_f}{P_i}\right)$
b. $\Delta S = nRT \ln\left(\frac{P_i}{P_f}\right)$
c. $\Delta S = nR \ln\left(\frac{P_f}{P_i}\right)$
d. $\Delta S = nR \ln\left(\frac{P_i}{P_f}\right)$

24. The correct thermodynamic conditions for the spontaneous reaction at all temperatures is: *(2016 - I)*

a. ΔH < 0 and ΔS < 0
b. ΔH < 0 and ΔS = 0
c. ΔH > 0 and ΔS < 0
d. ΔH < 0 and ΔS > 0

25. For the reaction, $X_2O_4(l) \rightarrow 2XO_2(g)$, ΔU = 2.1 kcal, ΔS = 20 cal K^{-1} at 300 K. Hence, ΔG is: *(2014)*

a. –2.7 kcal
b. 9.3 kcal
c. –9.3 kcal
d. 2.7 kcal

Clausius Clapeyron Equation

26. Consider the following liquid-vapour equilibrium.

Liquid $\rightleftharpoons$ Vapour. Which of the following relations is correct? *(2016 - I)*

a. $\frac{d\ln P}{dT^2} = \frac{-\Delta H_v}{T^2}$
b. $\frac{d\ln P}{dT} = \frac{-\Delta H_v}{RT^2}$
c. $\frac{d\ln G}{dT^2} = \frac{\Delta H_v}{RT^2}$
d. $\frac{d\ln P}{dT} = \frac{-\Delta H_v}{RT}$

Answer Key

1. (b)	**2.** (c)	**3.** (a)	**4.** (d)	**5.** (a)	**6.** (c)	**7.** (a)	**8.** (d)	**9.** (a)	**10.** (d)
11. (a)	**12.** (c)	**13.** (b)	**14.** (d)	**15.** (d)	**16.** (a)	**17.** (a)	**18.** (b)	**19.** (c)	**20.** (d)
21. (d)	**22.** (c)	**23.** (d)	**24.** (d)	**25.** (a)	**26.** (b)				